

System Setup aur Installation Guide (Roman Urdu)

SoloLLM project ki local setup workflow aur iske kaam karne ka mukammal tareeqa yahan bayaan kiya gaya hai. Is documentation ka maqsad yeh samjhana hai ke backend server aur frontend UI mil kar kaise aik local AI ecosystem banate hain.

Project Structure ka Overview

Project do main hisson mein taqseem hai: 1. **Backend (Python)**: Yeh logic, AI models (inference), database aur file storage ko handle karta hai. Port 8000 par run hota hai. 2. **Frontend (Next.js/React)**: Yeh user interface hai jahan se end-user app ke sath interact karta hai (chatting, documents upload, settings modification). Port 3000 par run hota hai.

1. Backend Setup Kaise Kaam Karta Hai?

- Virtual Environment (.venv)**: Python ki sab dependencies isolated environment (.venv) mein milti hain taake system-wide packages kharab na hon.
- start_backend.pyw**: Yeh ek special Python script hai jo .pyw extension ke sath hoti hai, matlab yeh bina terminal window (invisible) background mein open hoti hai. Yeh script pehle Virtual Environment ko activate karti hai, .env files parhti hai, aur phir uvicorn command ke zariye main.py ko run kar deti hai.
- requirements.txt**: Yeh file backend folder mein maujood hai jismein saare Python packages aur unke specific versions listed hain (jaise fastapi==0.115.6, uvicorn). Is file se `pip install -r requirements.txt` execute kar ke sab tools background mein install hotay hain.

2. Frontend Setup Kaise Kaam Karta Hai?

- Node.js aur npm**: Frontend Node.js par dependent hai.
- package.json**: Yeh file frontend folder ki bunyaad hai. Ismein sab framework versions (React 19, Next.js 16) aur unke scripts (dev, build, start) maujood hain.
- npm run dev**: Developers is command ko chala kar Next.js ka local server start karte hain. Yeh TypeScript files (.ts / .tsx) ko instantly JavaScript mein compile kar ke browser

mein hot-reload feature deti hai.

3. Scripts aur Autostart Mechanism

Project `scripts` folder mein kuch batch (`.bat`) files muqarrar hain jo app ki ease of use ke liye hain: * `install_autostart.bat`: Yeh script Windows ke Start-up folder mein backend ka shortcut daal deti hai. Jab aap PC on karte hain, toh background mein automatically backend server (LLM API) start ho jata hai, jis se aapko bar bar manually script nahi chalani parti. * `uninstall_autostart.bat`: Agar user chahta hai ke automatic start band ho jaye, toh yeh script us shortcut ko remove kar deti hai.

4. Background Downloads aur Dependencies Execution

1. **Ollama Integration**: System sab se pehle check karta hai ke system mein `ollama cli` (Command Line Interface) installed hai ya nahi. Agar nahi, toh yeh user ko guide karta hai ya automatically script ke zariye official source (`ollama.com`) se fetch karke setup karta hai. Uske baad hi models ko load kiya jata hai.
2. **Database Creation**: First time chalne par `aiosqlite` backend mein `solollm.db` (database file) automatically create kar leta hai aur tables (conversations, messages, settings) initialize karta hai.
3. **Vector Models**: Chunking aur RAG (documents padhne wala feature) ke liye `sentence-transformers` background mein models cache folder (e.g. `~/.cache/huggingface`) mein download kar leta hai.

Mukammal Cycle

Jab aap system start karte hain: 1. App backend background process ke tor par zinda ho jata hai. 2. Frontend Next.js server app ko render karta hai. 3. Jab aap type karke "Send" karte hain toh Frontend ek HTTP POST / SSE connect backend ko bhejta hai. 4. Backend database state update karta hai aur Ollama ko as proxy invoke kar ke LLM inference raw mode mein laata hai. 5. Response stream ho kar wapis apke screen par (MessageBubble) typewriter effect ke sath dikhaya jata hai!